IBM Qiskit Fall Fest 2026 Hackathon

MIT World Peace University, Pune, India

October 31, 2026

Introduction

Welcome to the Qiskit Fall Fest 2026 Hackathon! Your mission is to develop a novel project that gamifies or advances the field of Quantum Error Correction (QEC). You have 3 hours to complete two of the three tracks below. Use of online resources and AI is permitted. But remember that your project will be judged on integration of Qiskit Add-ons (10%), methodology (20%), conceptual breakthrough (30%), and your final presentation (40%).

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Track 1: Error Detection via Spacetime Codes

Challenge: Standard QEC codes detect errors in space (e.g., at a single moment in time across many qubits). Spacetime codes, however, spread the detection process itself across both space and time. This can reduce the hardware overhead, which is critical for near-term quantum devices.
Your task is to implement and analyze a spacetime code using the experimental 'qiskit-paulice' package.

1. **Circuit Design:** Construct a quantum circuit that performs a computation (your choice, e.g., a QAOA ansatz or a simple VQE circuit block) that includes a high density of T-gates.
2. **Spacetime Checks:** Instead of using dedicated spatial ancillas, embed 'Spacetime Checks' using the 'qiskit-paulice' operators. These should be configured as 'Controlled-Pauli-X/Y/Z' operations that are applied across specific localized spacetime coordinates, not just full-qubit-register time slices.
3. **Post-Selection Filter:** Implement a custom post-selection filter. This filter must process the ancilla readout syndromes to discard individual experimental shots that show evidence of error corruption.

Scientific Question to Address in Presentation: How did the placement of your 'Spacetime Checks' (measured in terms of their space-time separation) affect the fidelity of the logical state? Compare the post-selected fidelity of a Clifford-only circuit versus your T-doped circuit.

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Track 2: Advanced Error Mitigation with PNA, SLC, and NoiseLearnerV3

Challenge: Error mitigation techniques can be computationally expensive to run on full-scale devices. To scale, we must localize mitigation to the specific gates and qubits where errors are most likely to occur. This track challenges you to create a workflow that automatically composes three advanced Qiskit Add-ons.

1. **Pipeline Construction:** Build an automated workflow using Qiskit primitives that accepts a parameterized quantum circuit and a simulated noisy backend.
2. **Local Isolation:** Implement **SLC (Shaded Lightcones)**. Your script must identify the causal lightcone of specific entangling gates (e.g., a chain of CZ gates) and restrict the mitigation procedure to only this subset of qubits and operations.

3. **Dynamic Mitigation:** Apply **NoiseLearnerV3** to characterize the local noise parameters within the isolated lightcones, and then use **PNA (Propagated Noise Absorption)** to dynamically mitigate the effects of this noise as it propagates through the circuit layers.

Scientific Question to Address in Presentation: Quantify the trade-off between execution time cost and the final mitigated state fidelity as you vary the noise scaling factor (e.g., using a 'NoisyModel' with a variable error rate multiplier). How effective is SLC in reducing the overall mitigation overhead?

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Track 3: Tensor Network Surface Code Decoding & Reduction

Challenge: Tensor networks are powerful mathematical tools for representing and manipulating complex many-body quantum states. They are increasingly being used as sophisticated decoders for surface codes. This track asks you to implement a tensor-network-based decoder from scratch.

1. **Tensor Network Simulation:** Create a program to simulate a corrupted distance-3 surface code patch. Represent the state as a 2D network of local probability and check tensors.
2. **Network Reduction:** Write a tensor contraction script that systematically reduces this 2D network down to a single four-element array. Your goal is to perform this contraction with **minimal truncation error** (e.g., using approximate contraction methods).
3. **Fidelity Score:** Calculate the dominant logical correction operator from the reduced array. Compute a precise decoding fidelity score that reflects how accurately and quickly the logical state was protected.

Scientific Question to Address in Presentation: Introduce an artificial asymmetric erasure error model (where bit-flip errors are 10 times more likely than phase-flip errors) across the grid. How did your contraction strategy adapt to this, and how did the dynamic adjustment of the bond dimension truncation threshold impact the final logical fidelity?

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P.S. Remember, this hackathon is ultimately about your intuition, creative problem-solving, and how you approach quantum error correction—not just finding a single 'correct answer'! If you run into any technical issues, or have questions along the way, please don't hesitate to reach out to our volunteers or co-organizers. We're here to help!

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All the best!